

BMET3997 Physiological signal sensing and processing Major Project Submissions, Presentation, Report format and rubrics.



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We acknowledge the tradition of
custodianship and law of the Country on which
the University of Sydney campuses stand.
We pay our respects to those who have cared
and continue to care for Country.



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Submissions

- **Final submission must be emailed by Friday 5pm 5th June.**

Project Presentation (5%) plus Questions (5%)

All presentations will be held over 2 sessions on Monday 25th May. Session 1 in the lecture room. Session 2 is in in the lab room. Your team has been randomly allocated to a session.

- Your team has 6 minutes to present, followed by 4 minute of questions (strict timing!). If you go over time, you will be stopped and you will receive 0 marks for any presentation sections not covered!**

All team members must present.

All team members will be asked at least one question.

You must stay for the duration of your session. Anybody absent without special consideration or prior agreement with tutors and myself, will receive a mark of 0 for presentation.

Project Presentation times

Approx timing will be:

Lab, room J03.461 2-5pm	Lecture, J01.03.301, 10-12pm
Group 18: 2:10-2:20pm	Group 1: 10:10-10:20am
Group 6: 2:22-2:32pm	Group 2: 10:22-10:32am
Group 3: 2:34-2:44pm	Group 4: 10:34-10:44am
Group 16: 2:46-2:56pm	Group 13: 10:46-10:56am
Group 11: 2:58-3:08pm	Group 9: 10:58-11:08am
Group 7: 3:10-3:20pm	Group 20: 11:10-11:20am
Group 17: 3:22-3:32pm	Group 22: 11:22-11:32am
Group 14: 3:34-3:44pm	Group 12: 11:34-11:44am
Group 8: 3:46-3:56pm	Group 10: 11:46-11:56am
Group 5: 3:58-4:08pm	
Group 21: 4:10-4:20pm	
Group 15: 4:22-4:32pm	
Group 19: 4:34-4:46 am	

Project Presentation 5% [30 marks]

6 minutes presentation of your solution as a team (5% - team work)

Powerpoint slides only (recommend 6 slides)

– Topics for you to cover:

- Provide an overview diagram summarising your proposed approach and explain it. [5 marks]
- Describe detail on [15 marks]
 - QRS detector methods
 - HRV methods
- What performance your system achieves so far (see next slide for some suggestions on format presenting these results) [5 marks]
- Describe any issues you've had so far and your solutions. [5 marks]

Project Questions 5% [10 marks]

4 minutes(individual)

All group members will be asked at least one question. Please keep your answers short and to the point as there is approximately 1 minute each

Presentation/Final report: format for presenting performance results

	Your estimated performance using the training set examples					Independent Testing set performance				
Submission	1	2	3	4	5	1	2	3	4	5
Sensitivity (%)										
Positive predictivity (%)										
F1 score (%)										
$MAPE_{avgRR}(\%)$										
$MAPE_{sdRR}(\%)$										
$MAPE_{RMSSD}(\%)$										
$MAPE_{pNN50}(\%)$										
$MAPE_{LF}(\%)$										
$MAPE_{HF}(\%)$										
$MAPE_{LF_HFratio}(\%)$										
Average(MAPE)(%)										

Rubric – presentation (30 marks)

Criteria	HD (85-100%)	D(75-84%)	C(65-74%)	P(50-65%)	F(0-49%)	Marks
1. Overview Diagram & Explanation	Diagram is clear, logical, fully matches explanation; shows all key steps	Diagram mostly clear; minor gaps	Diagram and explanation partially clear	Diagram or explanation minimal	Missing or unclear	5
2. Details of Solution (QRS detector, HRV methods)	Full detail on all relevant methods, preprocessing, structure, thresholding	Mostly detailed; minor gaps	Some methods described; partial coverage	Minimal description	Missing or incorrect	15
3. Performance on Training & Testing Data	Clearly presents results for training and testing records; explains differences	Mostly clear; minor gaps	Some results presented; partial explanation	Minimal results; unclear explanation	Missing or incorrect	5
4. Issues Encountered & Solutions	Clearly identifies problems and well-reasoned solutions	Some issues and solutions discussed	Minimal issues or solutions	Very limited	Missing	5

Rubric – questions (10 marks)

Criteria	HD (85-100%)	D(75-84%)	C(65-74%)	P(50-65%)	F(0-49%)	Marks
Questions / Engagement	Fully answers questions in a concise fashion; demonstrates deep understanding	Answers questions with some understanding	Answers limited; basic understanding	Minimal engagement	Unable to respond	10

Final report (35%)

- **Due Tuesday 9th June, 11:59pm**
- **Individual report on major project**
 - 15-20 pages per student.
 - 35%
- **Please submit through Canvas**

Final report [100 marks]

- **Title page** [list all group members and project group number]
- **Abstract**
- **Table of Contents**
- **Group contribution**
- **Describe the training database [5 marks]. Also include information on**
 - Number of cases
 - Sampling rate, amplitude quantisation
- **Provide an overview diagram summarising your group's approach to QRS detection and HRV calculation. [10 marks]**
- **Provide performance details of all submissions (see following slides). Include performance of HRV parameters on training data [10 marks]**
- **Individual contribution**
- **Introduction (Describe what your role was)**
- **Describe what you developed (e.g QRS detector, HRV calculation)**
 - **Methods [40 marks]**
 - Detail the methods you used and justify why you uses these approaches
 - Other details you may include
 - For QRS detector:
 - » how did you detect noisy signal that was not further processed
 - » Did your algorithm perform secondary searching?
 - For HRV:
 - » How did select all the physiologically reasonable RR intervals?
 - » How did you process missing values?
 - » How did calculate spectrums so they could be averaged
 - **Testing [15 marks]**
 - describe the testing of any software modules you implemented and outcomes from the tests). Provide evidence of your testing.
- **Future improvements [10 marks]**
 - Describe how your module could be further improved
- **Performance mark [10 marks].**
 - See next slide. Same for all group members

Report Rubric

Section	HD (85–100%)	D (75–84%)	C (65–74%)	P (50–64%)	F (<50%)
Database Description (5 marks)	Detailed dataset summary including cases, sampling rate, quantisation; clearly identifies issues and limitations.	Clear description of dataset with most details and some issue discussion.	Adequate overview; minor details or issues missing.	Basic summary; limited awareness of data issues.	Incomplete or incorrect information; no understanding shown.
System Overview & Diagram (10 marks)	Clear, well-designed diagram; logical workflow and concise high-level explanation.	Good diagram and explanation; minor clarity gaps.	Understandable but missing key detail.	Minimal diagram or vague description.	No meaningful overview or incoherent explanation.
System Performance (10 marks)	Clear presentation of results (table); insightful comparison of training vs testing performance with valid interpretation.	Good presentation; identifies main differences.	Basic results shown; limited discussion.	Minimal or unclear evaluation.	No valid results or wrong interpretation.
Development Section - Methods (40 marks)	Thorough, well-justified technical development with evidence (figures, code, metrics); shows clear understanding and justification.	Solid technical work with justification and results.	Adequate explanation and some testing; limited depth.	Basic work with little justification or testing.	Poorly explained, incomplete, or incorrect technical content.
Development Section - Testing (15 marks)	Convincing evidence of testing of methods	Solid evidence of testing of methods.	Adequate evidence of testing of methods.	Basic evidence of testing of methods.	Little to no testing of the implemented methods.
Future Improvements (10 marks)	Realistic and technically sound improvement ideas showing reflection.	Good ideas; mostly feasible.	Some suggestions; limited insight.	Minimal or vague improvements.	No improvements or unrealistic suggestions.
Performance mark (10 marks)	See next slide				

Final report – Performance mark

- Performance of your system mark (out of 10). This mark will be calculated as follows.

F1-score (%) on independent test recordings	<98.00	98.00- 98.99	99.00- 99.49	99.50- 99.69	99.7+
Mark	1	2	3	4	5

+

avgMAPE score (%) on independent test recordings	>50.00	35.00- 49.99	20.0- 34.9	10.1- 19.9	<10.0
Mark	1	2	3	4	5



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